

100 Days of Azure DevOps

Day 17 Handout — Permissions are seatbelts — annoying until someone force-pushes main into the sun

2 - Azure Repos & Git Mastery · Personal lab only · Educational content · Not a sales pitch

Architecture A — identity map (who can touch which ref)

Identity	Typical Git rights	Lab setting
Project Collection Admin	Bypass almost everything	You only
Project Administrators	Repo create/delete, security	You
Contributors	Contribute (push to non-protected), PR	Default if a friend joins
Readers	Clone / pull	No push
<Repo> Bypass policies	Push to main ignoring PR	Deny for Contributors
Force push (rewrite)	git push --force on that ref	Deny on main for everyone but you-as-breakglass

Architecture B — fork vs branch permission (pick the contribution model)

Model	How code arrives	Use when
In-repo feature/*	Same repo, permission on refs	Trusted small team (this lab)
Fork + PR	Fork repo in Azure Repos or GitHub; PR across forks	Open source / untrusted contributors
GitHub mirror	Second remote; extra origin	Optional public notes later — not required
TFVC check-in	Centralized lock	Out of scope
Public project + Contributors	Anyone with link may leak	Keep azure-100-labs private
Bypass for 'just this once'	Direct push to main	Treat as an incident, not a workflow

Architecture C — pitfalls that look like 'Git is broken'

Pitfall	What you see	Fix
Force push allowed on main	git push --force origin main succeeds	Repos → Security → Deny Force push (main)
Bypass policies = true for Contributors	PR required checkbox is theatre	Deny Bypass policies on the ref
Fork of a work repo into personal	Employer code on your MSA	Never. Lab uses only your commits
Inherited allow + deny confusion	Effective = Allow when you thought Deny	Read the Effective column, not just the row you edited
Guest with Project Admin	They can rewrite security	Audit Groups; remove guests
Relying on .git/hooks to stop force push	Other clone still force-pushes	Server ACL + Day 19 policies

One-liner to remember

Permissions are seatbelts — annoying until the day someone force-pushes main into the sun.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal org Project settings only. Document the ACL. Invite nobody from work. Forks optional.

- 1 Project settings → Repositories → select azure-100-labs → Security.
- 2 Select Contributors. Find Force push. Set Deny (or confirm inherited Deny) for this repo / refs/heads/main if scoped.
- 3 Find Bypass policies when pushing / Bypass policies when completing pull requests. Deny for Contributors.
- 4 Confirm your user (as Project Admin) still can administer. Do not add a work AAD group.
- 5 Write docs/repo-permissions-day17.md: Contributors may contribute via PR; cannot force-push main; cannot bypass policies.
- 6 Optional: GitHub → new private repo → add remote github and push notes only. Never mirror a work repository.
- 7 Commit the markdown on feature/day17-permissions and open a PR. Do not screenshot employer orgs.

Notes — repo Security click path (no REST required today)

```
# Project Settings → Repositories → azure-100-labs → Security
# Contributors:
#   Force push = Deny
#   Bypass policies when pushing = Deny
#   Bypass policies when completing pull requests = Deny
# Readers: no Contribute
# Documented in docs/repo-permissions-day17.md
```

Done checklist

- Can explain fork workflow vs in-repo feature branches
- Contributors cannot force-push main (Deny)
- Bypass policies denied for Contributors
- docs/repo-permissions-day17.md committed on the personal repo

Tomorrow — Day 18

Migrating repos to Azure Repos